26) Let $A = \begin{bmatrix} 2 & 0 & 6 \\ -1 & 8 & 5 \\ 1 & -2 & 1 \end{bmatrix}$, let $b = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$, and let W be the set of all linear combinations of the columns of A.

a) Is b in W?

b) Show that the third column is in W.

$\boxed{\text{Soln:}}$ W is the set of all linear combinations of the columns of A, i.e. W is the span of the columns of A.

To check if b is in W, we must check if for some $x_1, x_2, x_3 \in \mathbb{R}$,

$$x_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 8 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$$

This is equivalent to checking if the augmented matrix $\begin{bmatrix} 2 & 0 & 6 & 10 \\ -1 & 8 & 5 & 3 \\ 1 & -2 & 1 & 3 \end{bmatrix}$ has a solution set.

$$\begin{bmatrix} 2 & 0 & 6 & 10 \\ -1 & 8 & 5 & 3 \\ 1 & -2 & 1 & 3 \end{bmatrix} \xrightarrow[\times \frac{1}{2}]{\text{Row1} = \text{Row1}} \begin{bmatrix} 1 & 0 & 3 & 5 \\ -1 & 8 & 5 & 3 \\ 1 & -2 & 1 & 3 \end{bmatrix} \xrightarrow[+\text{Row1}]{\text{Row2} = \text{Row2}} \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 8 & 8 & 8 \\ 1 & -2 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow[-\text{Row1}]{\text{Row3} = \text{Row3}} \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 8 & 8 & 8 \\ 0 & -2 & -2 & -2 \end{bmatrix} \xrightarrow[\times 1/8]{\text{Row2} = \text{Row2}} \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & -2 & -2 & -2 \end{bmatrix} \xrightarrow[\text{Row3} \times -\frac{1}{2}]{\text{Row3} =}$$

$$\begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow[-\text{Row2}]{\text{Row3} = \text{Row3}} \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

<u>Verification of whether the final matrix is in echelon form</u>

1. Row 1 and 2 are non-zero, but Row 3 is zero. ∴ The condition that all non-zero rows must be above zero rows is True.

2. Row 1 Leading element $= 1$ in Column 1

Row 2 Leading element $= 1$ in Column 2